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Lesson Notes

MODULE 7 | LESSON 3

MAGNIFYING AND FRICTIONAL FACTORS

Reading Time	70 minutes
Prior Knowledge	Volatility, Credit Risk, Financing, Volatility, Correlation, Basic pricing of stocks
Keywords	Difference types of leverage, Nonlinearity, Liquidity Characteristics, Regulation

In the previous lesson, we reviewed credit risk, financing, volatility, and correlation. In this lesson, we'll review four more factors: leverage and non-linearity; regulation and liquidity. Let's begin with the magnifying factors of leverage and non-linearity.

1. Magnifying-Related Factors

Leverage and non-linearity are magnifying-related factors. They help to amplify returns. Leverage increases the absolute value of returns by simply having a greater size of investment. Non-linearity can amplify returns by having an influence that exceeds linear increments. There are different methods by which leverage and non-linearity operate. Let's begin with leverage and see different contexts in which it is used.

2. Magnifying-Related Factors: Leverage

Leverage can refer to different concepts or have different contexts. Let's visit each in turn.

2.1 Leverage as Force

Sometimes, people use the term leverage to describe a force or an exertion. Consider our example of a seesaw (or using the British term, a teeter-totter). You could have an adult and child stand on opposite sides of the pivot point of a seesaw. Yet, if the child sits further away, the extra distance provides a leverage that allows the child's force to equal that of the adult. How does this example apply to bonds?

Consider a bond's duration. Suppose you treated each of the five cash flows of a two-year note (four coupon payments, one principal payment). Imagine you took a two-year note that has cash flows and a principal repayment. Now let's balance these along the seesaw. Bear in mind that the cash flows are

proportional to their discounted values (so the discounted cash flows are not the same size). How does this example apply to a bond?

Consider a bond's **Macaulay duration**, or simply *duration*, (for a bond that pays coupons semiannually) assuming that the relevant time is $t = 0$:

$$\text{MacD}_{t=0,r} = -(1+r) \frac{P'_0(r)}{P_0(r)} = \sum_{i=1}^N \frac{CF_i}{(1+r/2)^{t_i}} \times \frac{t_i}{P(r)}$$

where, as before, r is the yield to maturity, $P_0(r)$ = present value of all cash flows (i.e., the current price (current as in $t = 0$) of the bond), CF_i is the cash flow at time t_i , with t_i being the time of i -th cash flow.

This formula for duration was first proposed in 1938 by the Canadian economist Frederick Robertson Macaulay and perhaps the careful student might have noticed that we have (in the case of bonds that pay annual coupons):

$$\text{MacD} = (1+r) \times \text{MD}.$$

The duration is the weighted average time to receive the cash flows. In fact, you should notice that the formula can also be written as

$$\text{MacD}(P)_{t=0} = \sum_{i=1}^N t_i w_i$$

with

$$w_i = \frac{CF_i}{P(r)(1+r/2)^{t_i}}.$$

Let's make a few important considerations.

Macaulay duration is always less than or equal to the maturity and, more precisely, it is equal to the maturity of the bond only when the bond is a zero-coupon bond.

Note also that if we increase the coupon size, we're adding heavier members in the front, which then shortens the duration. If we increase the maturity, we're adding distance to the principal payment, and this increases the duration.

Duration represents an equivalence: a zero-coupon bond that behaves like the original bond. Suppose you were to replace all these cash flows with a single zero-coupon bond with a maturity equal to the duration.

Then, we have two bonds that would experience similar sensitivities when the yield curve changes in small, parallel shifts. This risk equivalence was the original intention of the Macaulay duration.

Even though the units of duration are time, duration was intended to measure risk, as we stated in Lesson 1 of this module already, and it says that we can replace a particular bond with a zero-coupon

bond whose maturity matches the bond's duration.

Of course, if the yield curve is shifting, if it's tilting as it is right now in many countries, we have an inverted yield curve. If it's going down, then that analogy doesn't hold, and that's why we can't just use duration to measure interest rate risk. Do not ever forget that all duration formulae presented so far require that interest rates along the yield curve move via parallel shifts or the formulae are inappropriate.

In this example, we're using duration as if it were a "lever," and this conveys the idea of leverage. To be clearer, suppose an investment manager expects interest rates to go down. We know that this would lead to an increase in the bond price. Then, the manager would want to lengthen duration; they extend the maturity to increase the leverage of the principal payment so it creates more leverage and moves the balance point out longer. On the other hand, if the manager expected interest rates to go up, then as this would lead to a decline in bond prices, the manager will try to manage this risk by lowering the duration.

2.2 Leverage as Financing

While we're discussing bonds, let's consider another way to think of leverage: borrowing money to increase the size of your investment. Most of the time when people think of leverage, they might think of borrowing cash. Suppose a trader has a strategy on which they make a ten percent return over a year. A \$100 investment would be worth \$110, earning a \$10 capital gain. This is just a straight cash investment. Imagine you can borrow \$50, combine that with \$50 of capital, and have that same \$100 invested. (For simplicity, let's assume 0% interest on borrowed funds). The \$100 investment is still worth \$110, earning a \$10 capital gain.

Now, when we compute the returns of each scenario, we notice that the first scenario returns 10%, while the second returns 20% (\$10 capital gain on a \$50 initial investment). The 10% return was doubled because the investment was leveraged 2-to-1. The two to one means you double the ten percent return you would have gotten, and that takes the ten percent return to twenty.

Recall the lesson in the module discussing leverage where you buy a home, say it costs \$200,000. Let's say that you put twenty percent down, and you borrowed eighty percent. In fractional terms, you're putting one fifth down (\$40,000) and financing 80% (\$160,000). Suppose the home appreciates ten percent to a value of \$220,000. You sell the home. You pay back the mortgage. Now you've made \$20,000 on an initial investment of \$40,000. That's a fifty percent return. That's because you were leveraged five to one. These results look very promising.

In fact, let's say that you're a hedge fund, and you're investing in some type of futures. Contract futures are among the most leveraged investments you can make. Suppose the fund invests \$5 down and borrows \$95. Suppose the investments makes 10%, or \$10. Calculating the return on the initial \$5 investment, we have a 200% return. Thanks to a leverage factor of 20-to-1.

The more leverage you have by borrowing funds, the more a return is amplified. Like volatility, leverage is agnostic to direction and works to improve gains or aggravate losses. Leverage is like time-lapse photography. Have you ever seen a nature show where the film is sped up so you can see the blooming of a flower over a period of seconds instead of weeks? That's what leverage does to returns; it speeds

them up. Leverage helps small returns become large returns. The news is great for gains but terrible for losses. A leverage of 10-to-1 means a ten percent return becomes a 100% gain, but a -10% return means a 100% loss! Worse, a 20-to-1 leverage means a 10% drop becomes a 200% loss. Such losses could wipe out a trader, portfolio, or business! Where leverage is involved, risk management needs to provide checks and balances.

2.3 Leverage through Derivatives

Options do not allow financing: their premiums must be paid in full. Yet, options offer a second way to enact leverage: through the multiplier factor. For example, one equity call option grants the holder the right to buy 100 shares of the stock. Suppose you bought one option. Using boundary conditions, we've seen that this costs less than 100 shares of stock. Now suppose the stock price increases. This results in a much larger increase in the call price. There is not a simple number (e.g., 100-fold) due to the non-linearity and time value, but it is much larger than the amount the stock increases. Thus, derivatives provide leverage through a multiplier factor. Depending on the derivative, one contract tends to control many shares of the underlying securities.

This multiplier factor is true for futures. For example, one future contract could also be for 1,000 barrels of oil. Moreover, futures have financing built in as well. Many futures contracts typically require only a 2% to 5% deposit. So a bullish investor would be able to take an even larger position by entering a futures contract, which uses both financing and a factor multiplier to access a greater number of underlying securities by using less capital.

2.4 Leverage and Credit Risk

Everything that we've described previously is affected by leverage. Let's go through them one by one. As an example with credit risk, suppose you apply for a mortgage. One of the factors discussed was the debt to equity. Prior to the Great Financial Crisis (GFC), it was relatively easy to get mortgages where the homeowner puts down less than 20%. Say the home is \$100, and the down payment is \$20. The problem is if the value of housing drops more than 20%, the homeowner is effectively underwater. Recall the option analogy, where the option is out of the money. We discussed this in Module 4. Prudent credit risk recognizes the leverage (debt-to-equity) of the borrower and ensures it can withstand temporary disruptions in income, housing corrections, or higher interest rates.

So if leverage is used and the result is a huge capital loss, there's a double whammy—the capital loss, and the need for payment of whatever funds were borrowed—the financing costs of using cash or other securities. Suppose the loss is enough to trigger a credit downgrade—after all, an extreme loss wipes out equity and increases debt. This downgrade could make the financing rate more expensive! For example, did you ever read the fine print on your credit card? If you miss a payment on a different card, and the credit card company finds out, they can raise the rate on their card. Why? They have learned the borrower is riskier, and the higher credit risk warrants the lender charging a higher rate.

Since the global financial crisis, many mortgage lending requirements have become stricter to prohibit debt-to-equity ratios that would be excessive. Fifteen years ago, the market was more competitive and relaxed about those; today, that has been made tighter. In part, these stricter controls helped prevent a

housing crisis during the COVID-19 pandemic. If you look at the housing crisis during the GFC, there weren't the safety mechanisms in place that are in place today.

2.5 Leverage and Volatility

Equity exchanges have avoided the problem of excessive leverage by setting upper limits on the amount that can be financed. Now let's add in volatility. If volatility is 20%, leverage can turn that into multiples. Imagine an investment that is leveraged 20-to-1. A 1% return looks like a 20% return. But if volatility is sufficiently high to produce a short-term return of 10% then, with simple multiplication, the return is 200%! Extreme values that can make or break the portfolio. Using more volatility or using more leverage will get a portfolio to more extreme values than without. From the outside, it may be difficult to distinguish whether a return was made using more volatility ("risk") or more leverage. One difference is that leverage will always incur some type of financing costs, which still exist even in the case of huge capital losses.

2.6 Leverage and Correlation

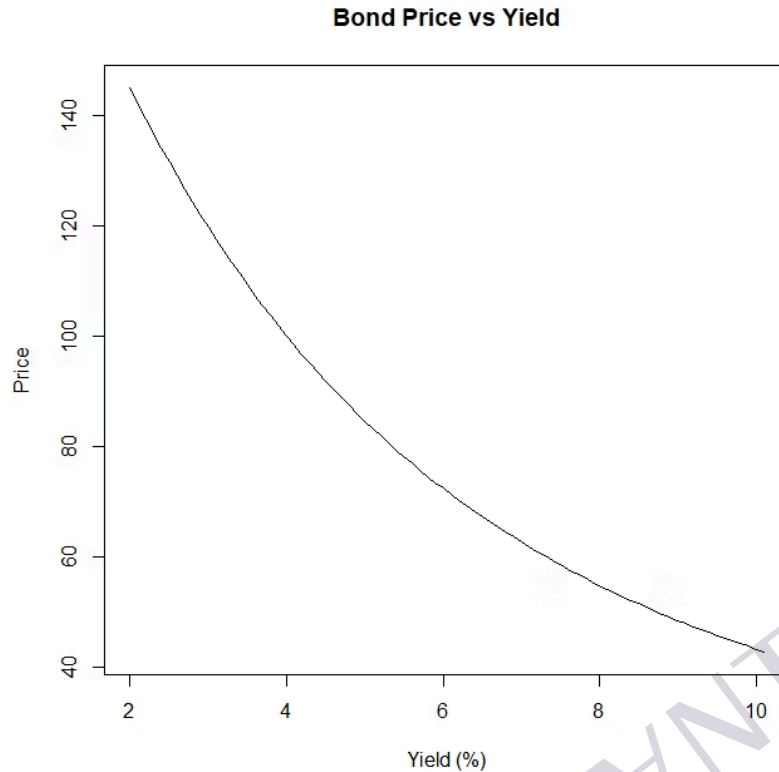
Correlation can amplify losses, particularly in credit risk. Consider the securitization examples you studied in previous modules. Higher default correlation causes the shape of the loss distribution to widen. Higher default correlation causes the securities in the pool to either survive together or default together. In a way, this leverages the returns to be extreme. Suppose default correlation increases to 1. Now we would have either complete default or no defaults. Indeed, here, default correlation is behaving like volatility. In your GWP3, think about the option position of the equity tranche.

3. Magnifying Related Factors: Non-Linearity

Oftentimes, we think of a system as linear. If we say that we invest four times the amount of capital, we should get four times the revenue. This illustration assumes we don't modify the amount of leverage. Sometimes, relationships between inputs and outputs are not linear. Let's discuss 4 examples of non-linearity.

3.1 Non-Linearity through Convexity

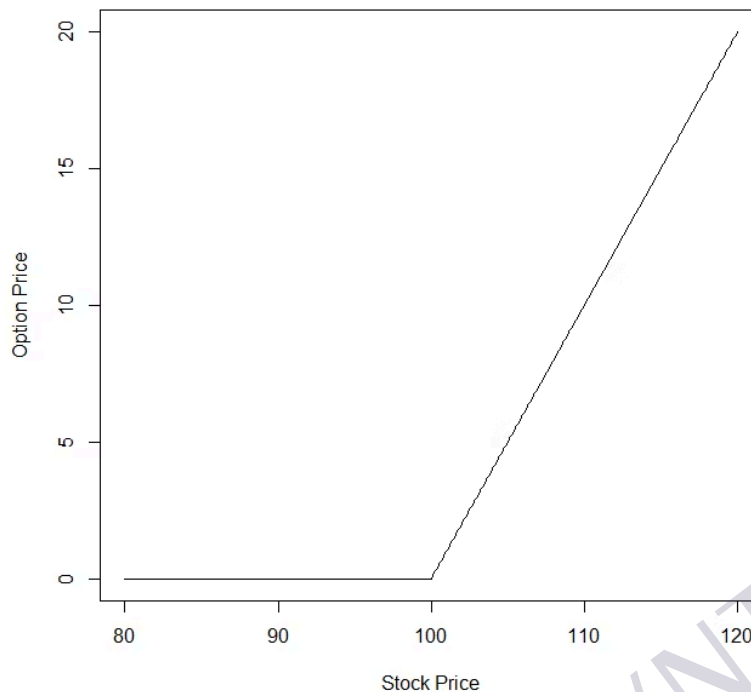
Let's graph a bond's price versus interest rates.



The relationship is both inverse and convex. Suppose we change the interest rates by 1bp. We get a price change called the Dollar Value of a Basis Point, known as the Dv01. Suppose we change the interest rates by 10bp. Is the price difference now just 10 times the Dv01? No, it is not due to the convex relationship. This is harder to deal with because when you're in the nonlinear world, this is an example that just has convexity. At Point A, yield changes produce relatively small price changes. At Point B, yield changes produce relatively larger price changes. Mathematically, convex relationships do not have a constant slope. In our bond price-interest rate graph, the slope changes from big negative values and then flattens as you increase the interest rate. The slope at point A is closer to 0 than the slope at point B. The non-linear world is harder, as we will see, because it is trickier to deal with maintaining the right ratio when hedging.

3.2 Non-linearity through Options 1: Payoff

Let's graph an option's payoff at expiration.



The payoff is piecewise linear: flat when the option is out of the money (OTM) and sloped when the option is in the money (ITM). The strike is where the curve changes from flat to sloped and is known as at the money (ATM). Although each portion is linear, the overall structure is non-linear. One way to think of non-linear is that the slope changes. At the strike, 100% of the change in slope occurs. Notice how the option payoff breaks symmetry. Putting a mirror on the strike would show that one side (OTM) is not a reflection of the other side (ITM). This means that call payoffs have unlimited upside but limited downside.

Suppose we are ITM. A small drop can keep the option ITM, but a large drop can make the option OTM. Therefore, changes in the underlying price stock do not necessarily result in proportional changes in the option payoff.

3.3 Non-Linearity through Options 2: Pricing

Recall that an option's payoff only applies at expiration. During the option's lifetime, an option has a price that we can model. Let's graph an option's price (using the Black-Scholes model you will learn later in the program). Note that the option's price is a nonlinear function of the underlying stock price. This is no longer piecewise linear but convex. Mathematically, notice how the curve is both smooth and continuous. This means the function will be differentiable.

3.4 Non-Linearity through Options 3: Sensitivities

The Greeks are the sensitivities of the option price to the underlying stock, or time, or the option hedge ratio or other terms. All these sensitivities are non-linear. Recall that the slope of bond prices to interest rates is also non-linear. Most products in fixed income and options have non-linear sensitivities.

One example of a security with a linear sensitivity is a stock. Financial practitioners often run the Capital Asset Pricing Model (CAPM), which produces a market sensitivity factor called beta. Beta is a

constant. A stock's beta doesn't change very often. When you take the derivative of a constant, you get zero. There is no second derivative. In the world of equities rather than get more mathematical, you tend to get a larger picture. Equity traders tend to specialize in sectors, such as technology. They understand management, earnings, market share, profitability ratios, and so on. It's quite a different skillset than for fixed income, FX, or options. You'll learn that options have Greek sensitivities—deltas, gammas, thetas, and vegas—that are nonlinear functions of the underlying. The result of this property is that options often need more frequent hedging than stocks or bonds do.

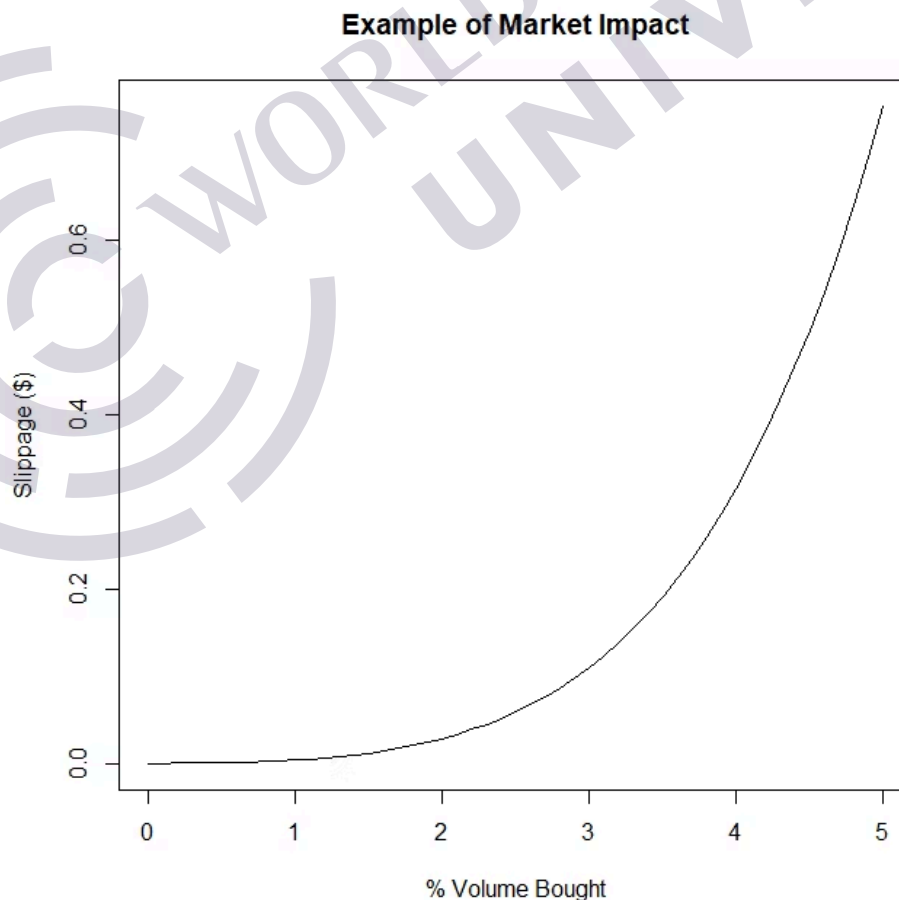
3.5 Non-Linearity through Variance

Recall from the module that discussed portfolio returns, the overall portfolio return is a linear combination of the individual returns, weighted by the proportion of each security's market value in the portfolio. Similarly, a portfolio of bonds would have a duration that is a weighted average of each individual bond's duration.

Variance is not linear. A portfolio of securities does NOT have a variance that equals the weighted average of each security's individual variance, except in the very special case where the correlations are equal to 1. The same is true for convexity. The same is true for option sensitivities. The disruption to this is correlation: in all but extreme cases (such as 1 or -1) we cannot simply use linear properties.

3.6 Non-Linearity through Liquidity

One final mention here is market impact. Consider a graph that illustrates market impact.



The y-axis shows slippage. Recall that slippage is defined as a price difference: the average price paid minus the price at the time the trade was entered. The x-axis shows trading volume. Trading volume is typically presented on a percent basis rather than outright basis. The graph conveys market impact. The greater the volume we buy, the greater the effect of market impact—typically, in a non-linear way. This relationship illustrates that a proportional increase in a volume traded could result in a disproportional slippage due to market impact.

The difficulty in detecting this type of non-linearity is that the effects can appear linear for small volumes. In other words, the effect may be proportional up to some threshold, and then becomes abruptly non-linear, much like an option payoff.

Market impact is a form of correlation. A large sell order influences buyers to lower their bid, creating slippage, which is the measure of market impact. Conversely, a large buy order will cause sellers to raise their ask prices, triggering market impact. Keep this in mind when you back-test trading strategies. When you use historical data to back-test a trading strategy, it did not contain the influence of your trades. If you are trading a small size, that is perfectly fine. If you are trading a size that creates market impact, then you are assuming that market impact is 0: turning a non-linear effect into a constant of 0.

3.7 Non-Linearity through Learning Functions

Many algorithms in machine learning have learning functions. The ones that tend to work best are the nonlinear ones. Of course, this will be addressed in Econometrics, Machine Learning, and Deep Learning in detail (like ReLu and Softmax). For now, we're just listing the wide variety of places where we see the need for nonlinear functions.

3.8 Non-Linearity Reviewed

The danger of assuming linearity when things are non-linear can be catastrophic. We'll say more about this in model failure. These challenges are not meant to discourage you from getting into financial engineering. They're really meant to show you what the pitfalls are along the way. Merely going through the motions of completing an assignment can get you through a course and perhaps the program, but thinking critically and creatively about how these models are addressing these challenges will develop your FE skills.

4. Frictional Risk Factors: Liquidity and Regulation

Liquidity and regulation are to financial markets what air resistance and friction are to physics. When you study physics, you initially ignore air-resistance and friction to make problems tractable, easier to solve, and simpler to interpret. Nevertheless, these frictions and resistances exist in the real world in the form of regulation and liquidity. In more advanced physics or engineering, they are accounted for when modeling physical systems. Likewise, liquidity and regulation are costs we must anticipate and calculate within models. Many books discuss the topics interchangeably, for example (Darrell Duffie's "Fragmenting Markets: Post-Crisis Bank Regulation and Financial Market Liquidity"). Let's begin with liquidity.

5. Liquidity

Let's consider different contexts in which liquidity is used in financial markets.

5.1 Liquidity as Solvency

At the individual or institutional level, one could assess liquidity by asking: Are there sufficient cash flows to pay the debt/bills? Is there enough cash flow to handle unexpected expenses that arise? These are questions of solvency. These questions pertain to individuals or companies that are faced with making debt payments over time.

For example, homeowners with a mortgage have an asset liability management problem. They have monthly income and expenses. Are they able to cover their monthly homeowner costs: mortgage, taxes, insurance, and utilities? Renters have the same problem. Are they able to cover their monthly rent and tenant-related expenses? Both mortgage lenders and landlords will assess the prospective homeowner or tenant by means of their ability to handle debt. Lenders and landlords assess this through requiring certain debt-to-equity ratios. For example, lenders may require no more than 40% of the income going toward the mortgage. Landlords may require an annual gross income that is 40 times the monthly rent. (In more lenient markets, perhaps landlords require less gross income, as credit risk may not be as much of an issue).

Suppose an applicant works on commission rather than a salary. The applicant has periods of time through the year where income decreases. If that applicant did not handle their cash flow well, they may find themselves behind on mortgage payments or rental payments. This unfortunate situation affected many borrowers and tenants during the COVID-19 pandemic. Many people lost their jobs and the sources of revenue. Nevertheless, the monthly liabilities of mortgage payments and rental payments remained. People sought relief. This type of protection will be discussed in regulation and crises.

5.2 Liquidity Based on Bid-Ask Spread

Aside from solvency, liquidity often refers to a security's value. Another way to define liquidity is how far from the true market value we can expect to trade a security. In a liquid market, a security trades very close to its true value. In an illiquid market, a security trades away from its true value.

The bid-ask spread is one of the most common ways to measure liquidity. From the sell-side (market maker) perspective, the bid-ask spread represents the profit margin the sell side extracts for providing a two-sided market. A market maker is willing to sell at the ask price and buy at the bid price. A wide bid-ask spread means that the market maker is willing to sell at a high price (ask price) or buy back the security at a low price (bid price). A tight bid-ask spread means that those 2 prices are closer to each other. Typically, the mid-quote is the average of the bid and ask price.

The bid-ask spread depends both on the security and on the type of market—exchange, OTC, retail—where the transaction occurs. For example, suppose you arrive in Tokyo, Japan, and want to convert your currency (say one million Nigerian naira) to Japanese yen. You're purchasing yen at the ask price. Then, you discover you must return home immediately. Now you're selling yen at the bid price. The transactions happened within such a short time, there was no change in prices. You'll wind up with

fewer naira than when you started because of the bid-ask spread. Instead of transacting at the airport, suppose you transacted with your global bank using their ATM. The exchange rates would likely be much better, meaning that the bid-ask spread was tighter. Therefore, the same FX transaction, naira to yen, would be more liquid electronically through a bank than through the airport kiosk. The kiosk has significant overhead (rent, labor) and a much more limited set of transactions than the bank and uses the bid-ask spread to increase the revenue to offset those relatively high expenses.

One huge advantage of exchanges is their ability to show market participants the limit order book. Suppose you are an aggressive trader and want to buy shares of stock. You could get the full ask size at the current ask price and second best ask size at the second best ask price, and so on. You could get as much as whatever the ask size is at that time. On the other hand you, suppose you place a limit order. Suppose you're only willing to pay a maximum of 99. Instead of taking supply from the market, you're adding to the supply of the market. The limit order book will enter your price and size (in the order in which the trade is received) until it is filled or canceled. Instead of taking liquidity from the market, you're willing to wait until someone is willing to sell to you at 99. There is a chance you do not get filled. Suppose the price only dropped to 99.05. Then, you would have protected your limit of 99, but you did not get filled. The lost opportunity may not have been worth it, if you feel the stock was undervalued 20%. A market order is different from a limit order because the market order gets filled, but perhaps at an unattractive price, whereas the limit order may not get filled, which is execution risk.

The bid-ask spread can also vary according to the time of day. Suppose you trade NFLX stock during active NYSE hours. The bid-ask spread will be relatively tight. Suppose you trade 12 hours later, when the NYSE is officially closed. There are only a few market makers providing liquidity overnight – but they do so at a cost that widens the bid-ask spread considerably. So the bid-ask spread depends on many factors: the security, the location, the time of day.

On the day before a holiday weekend begins, many market makers head out on vacation early. Not surprisingly, there are fewer skilled market makers. At these times, one can find the bid-ask spreads larger as this period behaves like an off-hours market.

5.3 Liquidity Based on Volume Traded

Another way to assess liquidity is to assess the volume that trades. On the crypto exchange called Binance, Bitcoin trades an enormous amount of volume. Typically, \$11 to \$12 billion of average daily volume is common. The S&P 500 is a common equity index containing the top 500 stocks by market value. Interestingly, the top 50 stocks (top 10%) on the list have much higher average daily volumes than the bottom 50 stocks (bottom 10%). So the top stocks are more liquid via their volume. The active (on-the-run) U.S. Treasury benchmark securities have significantly more average volume than that of the off-the-run securities. Moving to corporate bonds, some bonds may not trade for days or weeks. Recall that some portfolio managers hold bonds until maturity, redeeming their coupons along the way. Effectively, this means those particular bonds are not very liquid as measured by the volume. Finally, options have both a volume traded and new issuance. Recall that options are bilateral contracts, and so new ones can constantly be written as demand grows. Recent studies showed that some of the Bitcoin trading volume may be misleading (Paz).

5.4 Liquidity Based on Fungibility

Greater volumes can be traded when the items being traded are identical to each other. One share of Netflix common stock is like any other. Japanese yen are indistinguishable from each other. Five Bitcoins in your account are equivalent to 5 Bitcoins in another person's account. One hundred barrels of crude oil (with certain specifications) are like any other 100 barrels of crude oil.

Yet, the same may not be true for diamonds, which vary with several features (cut, color, clarity, and carat). The same is not true for homes because no two homes are exactly alike. Yes, they can have the same floor plan, materials, etc., but they cannot physically be in exactly the same space.

Fungible securities (stocks, dollar bills, Bitcoin) are interchangeable. Non-fungible securities (diamonds, homes) are not interchangeable. An extra level of due diligence is required at the individual security level when trading non-fungible securities. For example, it may be a buyer's market in real estate, but one still has to perform prudent valuation of a specific property. Generally speaking, markets with fungible products are more liquid than markets containing non-fungible securities. Examining the idiosyncrasies of non-fungible assets takes time. Homes take longer to examine, finance, and close than stocks. You could buy and sell stocks several times a day. If you're a high frequency strategist, you could do that thousands of times a day. It would take weeks or months to flip residential or commercial real estate because of the paperwork and complexities involved.

5.5 Liquidity Based on Time to Trade

Liquidity is also measured by the time it takes to get a trade executed. During active NYSE hours, trading stocks is quick. The same is true for stocks on NASDAQ. Now suppose you want to trade real estate. You could trade Real Estate Investment Trusts (REITs) on one of those exchanges very quickly. However, if you were to purchase a residential property, that would take weeks. The process is time-consuming, and depends on various entities (mortgage lenders, seller, realtors, appraisers, inspectors, local government officials) and is not intended to occur within the same day. This process is one of several reasons why real estate is less liquid than equities. (We've covered the other reasons—not exchange traded, not as much volume, lack of fungibility).

5.6 Liquidity Based on Market Impact

Liquidity is also measured by market impact. Market impact refers to the influence trades have on the limit order book. Suppose you're an aggressive buyer of Twitter (hypothetical, of course, because unfortunately Elon Musk beat you to it). As you put bids in to buy outstanding shares, you'll drive the stock price up. So that's what's known as market impact. The only chance you'd have to purchase the company is to pick a level that exceeds the current stock price.

Market impact reflects an increase in the price you will pay for a security when you buy that security. Let's say that when you start buying, you increase the price because your action in the market is correlated. If you keep putting bids in to buy, then your action is going to increase the value simply because other market participants, the market makers who are selling will be motivated to increase the price, thinking that you'll be willing to pay the higher price. There's a price increase as well as a time that it needs to bounce back. Market impact is like the rebound of a spring. A liquid security has a spring that reverts quickly. Then, you're stretching the spring a great deal, so you increase the price quite a bit from the impact, and then it's also slow to return to its equilibrium position. If something is

very liquid, which means it has little market impact, there will not be a big price increase, and it will take a very short amount of time to return to its original value.

Recall from the nonlinear section the discussion of market impact. A profitable trading strategy can turn into an unprofitable one. The expected alpha of the trade may be eroded by the market impact faced in each direction: facing slippage on entry by paying more than the current ask price and facing slippage on exit by selling for less than the current bid price. To avoid becoming predictable in your strategy, execution algorithms will slice and dice large orders to avoid detection from algorithms of the intent of the large buy order.

5.7 Liquidity Affects Credit Risk

When it is time to pay a debt, is there enough cash flow to pay the liabilities? This question is the core problem of asset liability management. From a solvency perspective, liquidity is important to assess credit risk. Imagine that a company has no debt, no cash flow requirements (salaries, rents, expenses, etc.). In this unusual case, liquidity is not an issue because there are no impending liabilities. Indeed, many growth companies are in a phase where they have pools of capital (venture capital, private equity, angel investors), are not issuing debt, and may not be expected to be profitable during a ramp-up phase. It often takes years to build a factory, patent a drug, or start a brand that becomes a profitable company. Expenses along the way create possibilities for the firm to default. Oftentimes, these expenses or negative profit margins create a burn rate that requires startups receive additional funding to keep going. The real test of liquidity is really to face liabilities, and the inability to do so is certainly a form of our credit.

5.8 Liquidity Review

In short, measures of bid-ask spread, volume, time to trade, and market impact are all considered when executing trades. Bid-ask spread is the most popular because it is easier to observe, and acts like a cost (e.g., 20 basis points). Suppose you buy a stock on an exchange at the ask price. The prices don't change. Then, minutes later, you decide to sell the stock. You sell at the bid price. Roundtrip, you paid one bid-ask spread. Some students think you pay it twice, but remember that each trade is paying about half the spread (ask price minus bid price going in, and bid price minus bid price going out). In some markets, the granularity between prices is quite small—one cent in stocks. In other markets, the intervals can be much further apart (e.g., \$32.50 in treasury markets).

5.9 Liquidity as Solvency Revisited: The Banking Level

We discussed liquidity risk at the individual or company level: will an entity have enough funds to pay its debt? Just like corporations, banks should have sufficient collateral or cash reserves to weather bad storms. While a bankrupt corporation is unfortunate as it causes job losses and regional economic decline, a bank declaring bankruptcy is worse. It erodes trust in the financial system, and possibly affects all the banks in the industry—both those with which it engaged in transactions (i.e., interest rate swaps) and even banks with which it did not because of the interconnected nature of the global financial industry. This notion of enforcing bank liquidity brings us to our next challenge: regulation.

6. Regulation

Regulation exists for many reasons: people or companies over-speculate, behave unethically, and behave unintentionally. The need to produce revenue may cause an excessive exposure of risk. Often, leverage and non-linearity contribute to this tremendously. Anyone who is OTM—a trader or portfolio manager who is down on the year, a CEO of a company that will have no income for a quarter—is incentivized to take leveraged risks. Big wins will put them deep ITM, with continued employment, big bonuses, and stock option valuations. Big losses, on the other hand, may not put them OTM, but their firm. If a trader loses $\$X$ for the year, the firm can still survive, but the trader may lose their job. If a trader leverages to try to make $\$9X$ or lose $\$10X$, they are no worse off for the loss but have a potentially big gain. Unfortunately, the trader's incentives are not aligned with those of the firm.

Let's divide our study of regulation into several groups. Let's start with regulation at the asset level.

6.1 Regulation at the Asset Level

Consider a firm that wants to exit a portfolio of mortgage-backed securities. As you recall, MBSs do not trade on exchanges but OTC. Whereas exchanges would have publicly transparent pricing, OTC is less transparent. Instead of prices, there may be indications. There are likely to be delays as well.

The regulation known as TRACE (trade reporting and compliance engine) requires that they report the trades within 15 minutes. TRACE reporting applies to Treasury bonds, agency bonds, corporate bonds, MBSs, and other asset classes. Firms must invest heavily in technological infrastructure to be able to capture and report the details of the trade through the proper channels in a timely manner. Failure to do so results in TRACE violations—fines paid for not providing a transparent market. TRACE provides information to potential buyers and sellers: this includes the security, the price, the size, the trade date and time, and the type of trade that was done. Suppose you sold a particular MBS. Within 15 minutes, the details of the trade are posted. This information provides guidance in price discovery since it is not exchange traded.

In addition to prices, you may also see the details of complex products like securitized portfolios. Consider an MBS. What type of credit ratings does the collateral have? How do the tranches work? What are the required product disclosures? Regulation requires a level of product transparency to ensure investors can perform due diligence to determine their rewards and risks. Through MBS documentation, you can understand the type of products that go in there, how the cash flows are distributed to service providers and investors and different tranche holders, and so forth.

6.2 Regulation at the Individual/Company Level: Homeowner, Borrower, CEO, CFO, Hedge Fund, Bank

Over time, increased regulatory requirements applied to individuals like brokers and other fiduciaries to avoid conflicts of interest and outright criminal activity. Regulation helps to protect the consumer, typically a retail or institutional investor. For example, communication must be recorded. Several banks in the U.S. were fined in 2022 for failing to record employee communications on social media platforms (CNBC). Within financial firms, all communication today is likely recorded: emails, texts, chats, posts, etc. Brokers who lie to customers are not performing their fiduciary responsibility. In the most egregious cases, some of the brokers went to prison (you'll read more about this in the next lesson).

An individual failing is bad, but that is not necessarily debilitating for an economy. In other words, if a bank issued a loan to a homeowner, and unfortunately, the homeowner defaults, it's not a disastrous

event for the bank. Enron was a company in the United States known for one of the largest accounting scandals. Company executives submitted fraudulent accounting reports. While there were many examples of those accounting scandals, Enron became the most publicized case. Traders, portfolio managers, and even employees were duped into thinking the company was very profitable when it was nowhere near that.

These accounting scandals motivated new regulation: Sarbanes-Oxley. This law punishes firms if they file fraudulent accounting reports and holds their senior executives criminally liable if they sign off on those documents. That criminal liability means that, in addition to fines paid by the company, the CFO or CEO who knowingly signs the audited financial statements and pretends like the company is profitable will pay fines, go to jail, and get banned from the industry. This regulation discourages would-be criminals from touting fraudulent financials to attempt to overvalue a company.

6.3 Regulation at the Exchange/Sector/Regional Level: Everything Else

As these lesson notes are being written, the FTX scandal has emerged. There are many transactions that remain doubtful and seemed to avoid any scrutiny, such as security lending and oversight. Compare that to equity exchanges, which have been around for hundreds of years and have had some form regulation for over 100 years. This history of regulation doesn't prevent failures and crises, but many existing controls help to limit the disaster. For example, we'll see the notion of exchange circuit breakers to help avoid panics turning into crashes.

On the other hand, nascent exchanges like FTX have shown to have so little regulation that actions such as financing and securities lending are unchecked compared to similar behavior in more established exchanges.

6.4 Regulation at the Global Level: SIFI (Systemically Important Financial Institution)/Global Economic Level

Finance is one of the most interconnected industries. If you consider the 15 or 20 largest financial institutions in the world, many do business with each other—through security lending, short-term borrowing, interest rate swaps, FX swaps, credit default swaps, and other transactions. The failure of one could cause widespread damage throughout the industry.

Let's suppose that Bank A trades with Bank B: the products include IR swaps, FX swaps, and CDSs. All these swaps trade OTC. Now let's say that Bank B also trades with Bank C, and they have similar products between them.

Now let's suppose that Bank A defaults. Clearly, this affects Bank B. Perhaps Bank B was paying fixed to A and receiving floating from A—when the floating rate was substantially higher! Now that cash flow is gone. In turn, suppose Bank B paid floating to C and received fixed. The cash flow from A formed a hedge that is no longer in place. This puts Bank B in a compromised situation, which now poses an additional credit risk to Bank C. In turn, every bank in the industry can have the same affect. Initially, Bank C may seem indifferent to Bank A's situation because the two banks have no direct transactions. Whether direct or indirect, an important global bank's failure threatens the entire financial industry. The issue is how widespread and significant a financial institution is.

These banks are known as SIFIs: systemically important financial institutions. We don't want the SIFI to fail. Prior to the GFC, these banks operated under a free option (our nonlinearity problem). If the banks took excessive risks that led to profits, they continued to exist, paid bonuses to management, kept expanding, etc. If those risks led to extreme losses (as they did during the GFC), the banks looked for a bailout using the popular catchphrase "We're too big to fail." In many ways, this was true. There are several lessons to be learned.

First, regulation attempts to identify and dismantle the free option clause. Without the free option, banks will take accountability for their own losses and so will behave more prudently. This idea will require firms to comply periodically with a set of well-defined stress tests.

Regulation created guidelines for stress testing that SIFIs need to perform annually. If SIFIs fail these tests, then they're considered to be out of compliance. In that case, other banks are discouraged from doing business with them. It's almost as if they lose a license to do business because they lose a status of trust. These stress tests address issues such as, "Is there enough cash to survive the next thirty days or the next year?" or "Could the bank withstand a 20% downturn in the equity markets?" There's stability ratios and leverage ratios, and you'll see this world of stress testing well down the road in Risk Management.

6.5 Regulation and Liquidity

Even SIFIs need to be liquid. If too many funds are being lent out or tied up in investments, the bank may have insufficient reserves set aside for unexpected losses, as revealed through rigorous stress tests. Homeowners with a mortgage only face this test at the time they apply for their mortgage. Over the life of a 30-year mortgage, they do not need to pass additional tests. Their ability to pay the monthly mortgage suffices.

These SIFIs need to have collateral, cushions, and reserve accounts set aside. These regulations make banking a tough business: a delicate balance of investing and lending to be competitive and create revenue and saving to provide for a rainy day. Banks not only compete with each other but also with FinTechs, disruptors to their industries, and bottom lines that significantly affect their profit margins.

In addition, most financial institutions have to maintain records and compliance for anti-money laundering (AML), Know-Your-Customer (KYC), disclosing fees, and more. Regulation exists to add quality controls to lending. Suppose a lender and borrower can agree on mortgage terms: the lender is satisfied with debt ratios, and the borrower takes responsibility for the payments. There are still industry requirements that would prevent certain mortgages today; these requirements were not in place pre-GFC. Still, these regulatory requirements are meant to minimize future mortgage crises. These requirements require more documentation rather than NINA (No Income No Assets) loans. They also require a greater down payment (e.g., 20%) to provide a cushion against a housing market downturn. Both of these help mitigate credit risk. That means that banks don't have to become the owners of homes that are underwater. This is how some of the regulation worked at the individual or company level.

The documentation and the process of going through a mortgage is highly scrutinized now, and that helped to minimize the number of defaults. The COVID-19 pandemic has been a stress test of sorts for

the mortgage market, and there was nowhere near the number of defaults at the time of the pandemic that we had during the Global Financial Crisis, so many of the regulations that were in place helped to avert that crisis.

7. Conclusion

In the previous lesson and in this one, we have covered eight of the challenges that intersect in most financial engineering problems. The last two require a very big picture: model failure and crises. To understand model failure, we'll want to use data in our models, so we postpone the last two challenges until the Financial Data course. Model failure and crises become the key areas for us to apply critical thinking. Finance is not physics, and the financial laws are not immutable. In our next lesson, we expand upon the cornerstone of financial markets—the ethical behavior of its participants.

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